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**CMT426**

**BUSINESS INTELLIGENCE AND ANALYTICS**

Assignment 1

Group 35

| Name                                | Matric. No | E-mail                       |
|-------------------------------------|------------|------------------------------|
| Siti Anisah binti Noor Jamin        | 160408     | anisahjamin03@student.usm.my |
| Nur Fatin Aina binti Moh Amir       | 160353     | nurfatinaina@student.usm.my  |
| Nurin Farah Izzati binti Mohd Rusdi | 160406     | inurin32@student.usm.my      |
| Siti Nawwarah binti Mohd Abd Hadi   | 160304     | nawwarah@student.usm.my      |
| Siti Nur Izzah binti Abdul Rahman   | 160302     | izzahrahman@student.usm.my   |

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## 1.0 Organisation & Problem Context

### 1.1 Overview of Organization

In this project, we selected Food & Beverages that focusing on a coffee retail business. The organisation operates as a coffee retailer named Maven Coffee Co. that sells various coffee beverages and related products to customers across different locations and periods. As competition in the food and beverage industry continues to increase, businesses need to analyse their customer purchasing behaviour, product performance and sales trends to make effective business decisions.

Hence, the organisation requires data driven insight to support strategic decision-making related to branch management, product focus and operational planning.

### 1.2 Decision-Making Problem

Based on the transaction data, the primary decision-making problem faced by the coffee shop business is:

**How should Maven Coffee Co. allocate its product focus, branch attention and operational strategy to improve overall sales performance?**

This business operates across multiple branches and offers a variety of coffee products to customers. However, differences in branch performance, customer purchasing behaviour and the product demand across each branch may affect the overall profitability and operational effectiveness. Thus, without the proper data analysis, the company may face difficulties in identifying high-performance branches, understanding the customer buying patterns and determining which products contributes the most sales revenue.

Therefore, the organization requires data-driven insight to support strategic decision related to branch management, product focus and the operational planning.

### 1.3 Analytical Objectives

The analytical objectives of this project are:

1. To evaluate and compare branch performance based on total revenue and transaction volume across three store locations.  
This objective aims to identify the best performing and underperforming branches to support management decision making and resource allocation.
2. To identify customer purchasing trends across different dates and times for better operational decision support.  
This objective helps to understand peak purchasing periods and customer buying behaviour to improve operational planning and marketing strategies.
3. To analyse the sales contribution of each product category, product type and product detail.  
This objective aims to determine which product contribute the most to overall sales performance and identify product with lower demand.

## 2.0 Data Acquisition & Understanding

Dataset URL: <https://www.kaggle.com/datasets/ahmedabbas757/coffee-sales>

The dataset used in this analysis was obtained from Kaggle. This dataset contains transactional sales records from a fictitious coffee shop named Maven Coffee Co. operating out of three New York City locations, Hell's Kitchen, Astoria and Lower Manhattan.

### 2.1 Dataset Description

The dataset is a structured transactional dataset presented in tabular format, where each row represents an individual product purchase within a transaction.

The main attributes used in this study are as follows:

1. Transaction ID – a unique identifier for each transaction
2. Transaction Date – the date on which the purchase was made
3. Transaction Time – the time at which the purchase was made
4. Transaction Qty – the number of units purchased
5. Store ID – a unique identifier for each store
6. Store Location – the branch where the transaction took place
7. Product ID – a unique identifier for each product
8. Unit Price – the selling price per unit
9. Product Category – the general category of the product, such as Coffee or Tea
10. Product Type – a more specific classification of the product
11. Product Detail – the detailed name or description of the product

The dataset contains 149,116 transaction records covering the period from 1 January 2023 to 30 June 2023. It includes sales data from 3 store locations, 9 product categories, 29 product types and 80 product details.

#### **Relevance to the Problem**

This dataset is relevant to the business problem because it provides detailed sales information for analysing product performance, branch performance and customer purchasing patterns. The transaction date and time help identify sales trends and peak hours, while store location supports branch comparison. Product category, product type, product detail, quantity sold and unit price are also useful for evaluating which products contribute most to sales performance. Therefore, the dataset is suitable for supporting decision-making related to product focus, branch management and operational planning.

## 2.2 Data Cleaning

Basic data cleaning was carried out before analysis to ensure the dataset was suitable for Tableau.

- Checked for missing values in the main variables – no missing values were detected.
- Checked for duplicate records – no duplicate entries were found.
- Verified the data types of all variables.
- Set Transaction Date as a date field, Transaction Qty and Unit Price as numeric fields, and Store Location, Product Category, Product Type, and Product Detail as categorical variables.
- Reviewed Transaction Time for time-based analysis.

These steps ensured that the dataset was ready for sales, product, store and time-based analysis.

## 2.3 Summary Statistic

| Item                         | Summary                                                                                                 |
|------------------------------|---------------------------------------------------------------------------------------------------------|
| Total transaction records    | 149,116                                                                                                 |
| Date range                   | 1 January 2023 – 30 June 2023                                                                           |
| Number of store locations    | 3                                                                                                       |
| Store locations              | Astoria, Hell's Kitchen, Lower Manhattan                                                                |
| Number of product categories | 9                                                                                                       |
| Product categories           | Bakery, Branded, Coffee, Coffee beans, Drinking Chocolate, Flavours, Loose Tea, Packaged Chocolate, Tea |
| Number of product types      | 29                                                                                                      |
| Number of product details    | 80                                                                                                      |
| Transaction time range       | 6.00 a.m. – 8.59 p.m.                                                                                   |
| Transaction quantity range   | 1 – 8 units                                                                                             |
| Unit price range (USD)       | 0.80 – 45.00                                                                                            |

## 2.4 Key Variables for Analysis

The key variables used in this study are:

- Transaction Date and Transaction Time to identify sales trends and peak hours
- Store Location to compare branch performance
- Product Category, Product Type, and Product Detail to evaluate product performance
- Transaction Qty and Unit Price to measure sales volume and price level
- Revenue to measure sales value

These variables are important because they directly support the analysis needed to answer the business problem.

### 3.0 Tableau Dashboard Development

#### 3.1 Dashboard 1

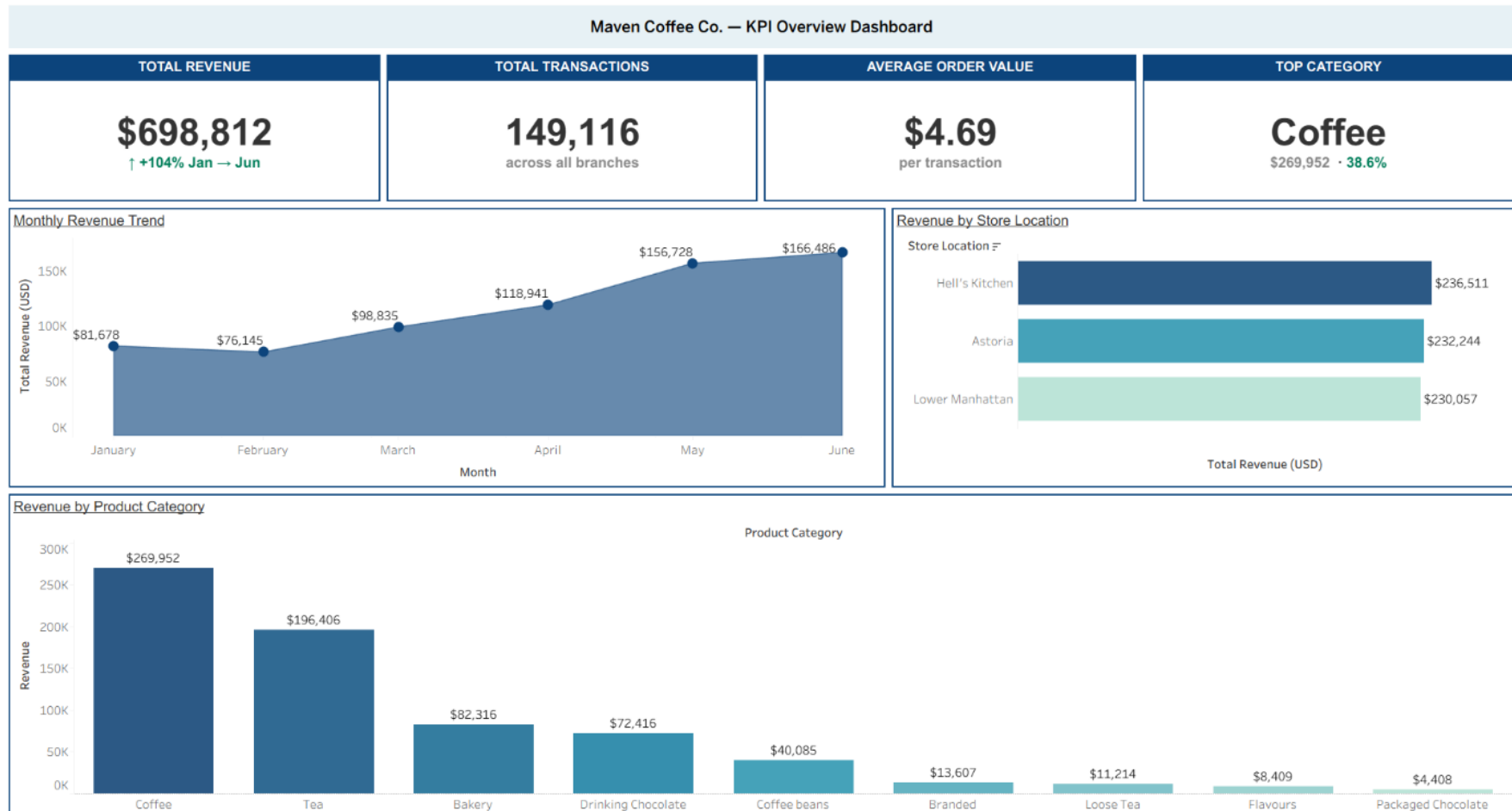


Figure 1: Dashboard 1 - KPI Overview

In Figure 1, Dashboard 1 presents a comprehensive overview of Maven Coffee Co.'s business performance across a six months period from January to June in 2023. It consists of four key analytical components including score cards, a monthly revenue trend, store location performance and product category breakdown. It serves as a centralised decision support tool which enables the management to monitor key performance indicators and make strategic decisions across products, locations and time periods.

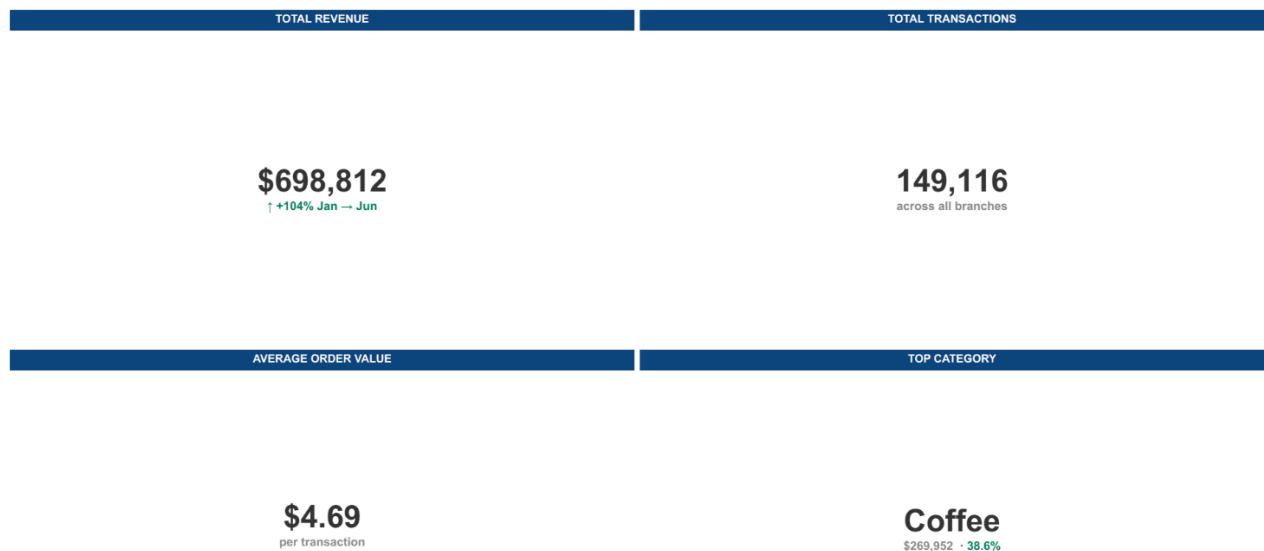


Figure 2: KPI Score Cards

In Figure 2, the score cards present four high-level business metrics in a summary format including:

- **Total Revenue (\$698,812):** represents the total income generated across all store locations from January to June which reflects a significant growth of 104% over the period
- **Total Transactions (149,116):** indicates the total number of purchases made across all store locations
- **Average Order Value (\$4.69):** reflects the typical amount customer spent per transactions
- **Top Category (Coffee):** identifies Coffee as the highest performing product category

These indicators provide management with an immediate overview of overall business performance without requiring deeper analysis of individual charts.

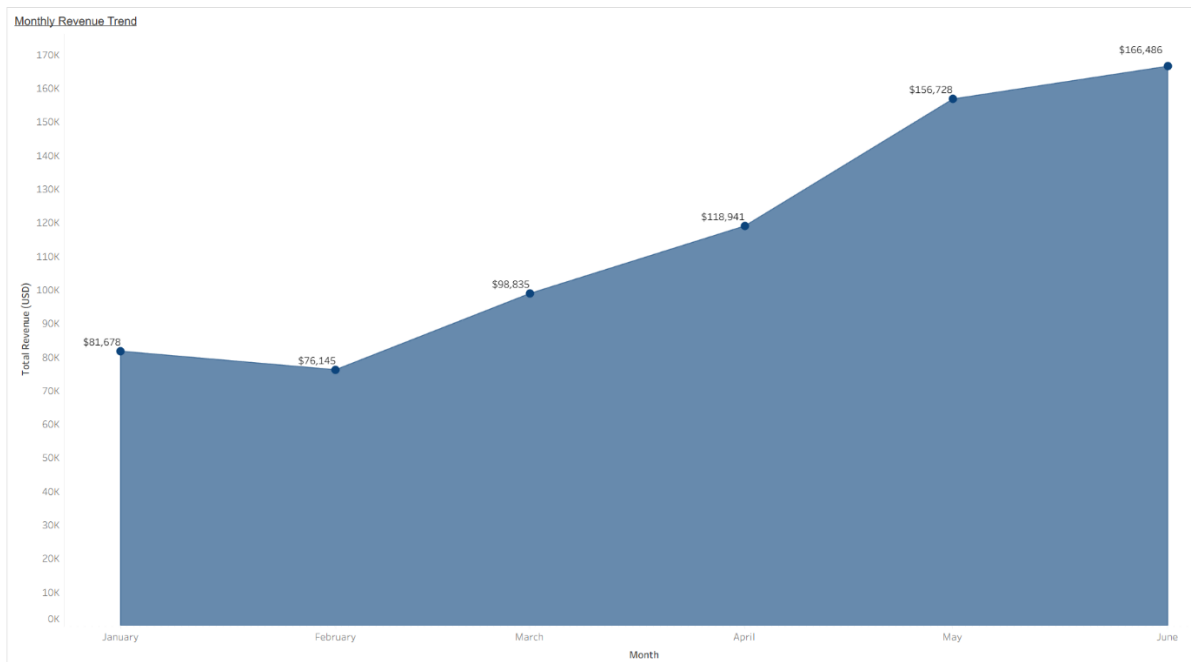


Figure 3: Monthly Revenue Trend

In Figure 3, the area line chart illustrates the progression of total revenue on month-by-month basis from January to June in 2023. There is a notable decline in February (\$76,145) and followed by a consistent upward trajectory peaking at \$166,486 in June. This helps management to monitor growth momentum and detect any periods of underperformed that may require strategic plan.

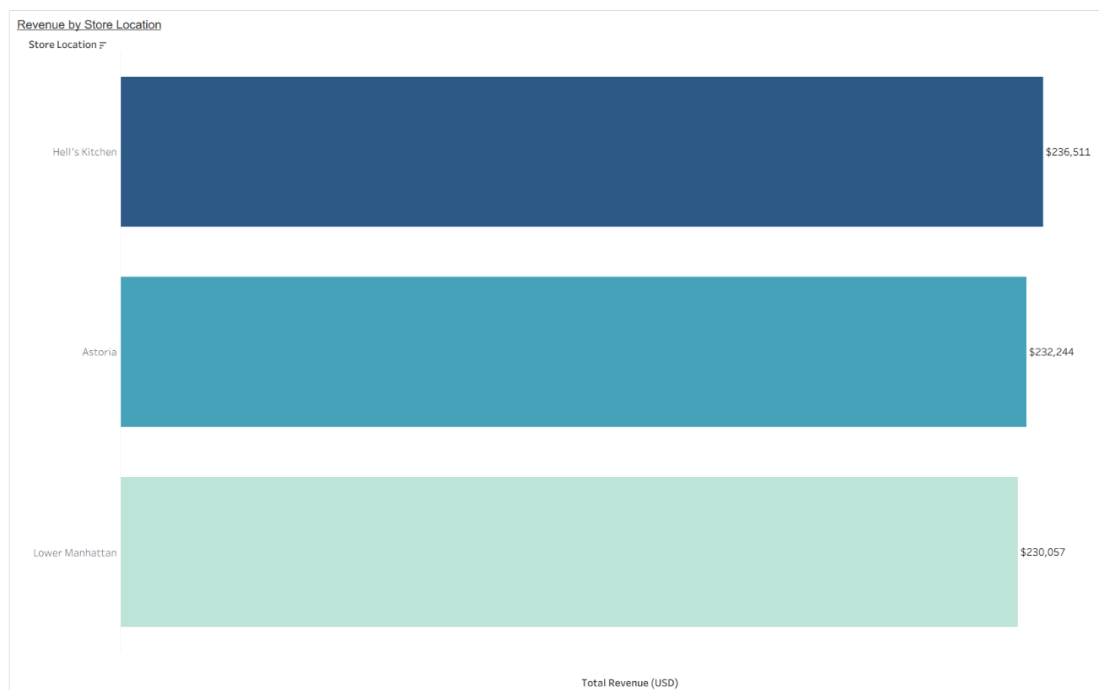
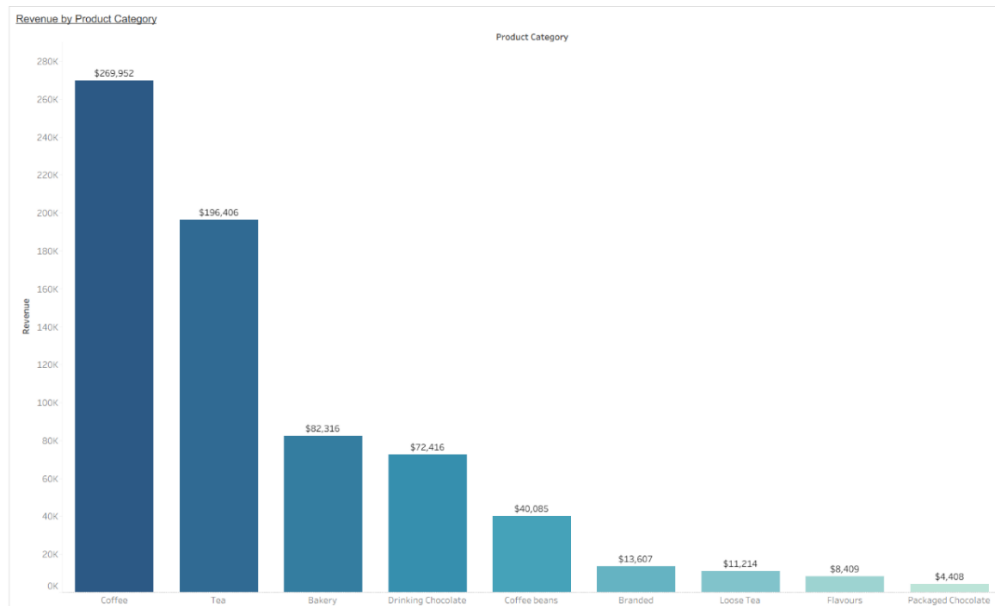


Figure 4: Revenue by Store Location

In Figure 4, the bar chart compares cumulative revenue generated across the three branch locations which are Hell's Kitchen, Astoria and Lower Manhattan. It is important as it provides a location-based performance comparison which allows the top management to assess branch-level contribution to overall revenue and determine whether any location requires additional support.





*Figure 5: Revenue by Product Category*

In Figure 5, the vertical bar chart ranks all nine product categories in descending order of revenue generated. It highlights the most commercially valuable product line and identifies underperforming categories which helps the management with informed decisions based on inventory management and promotional strategies.

### 3.2 Dashboard 2

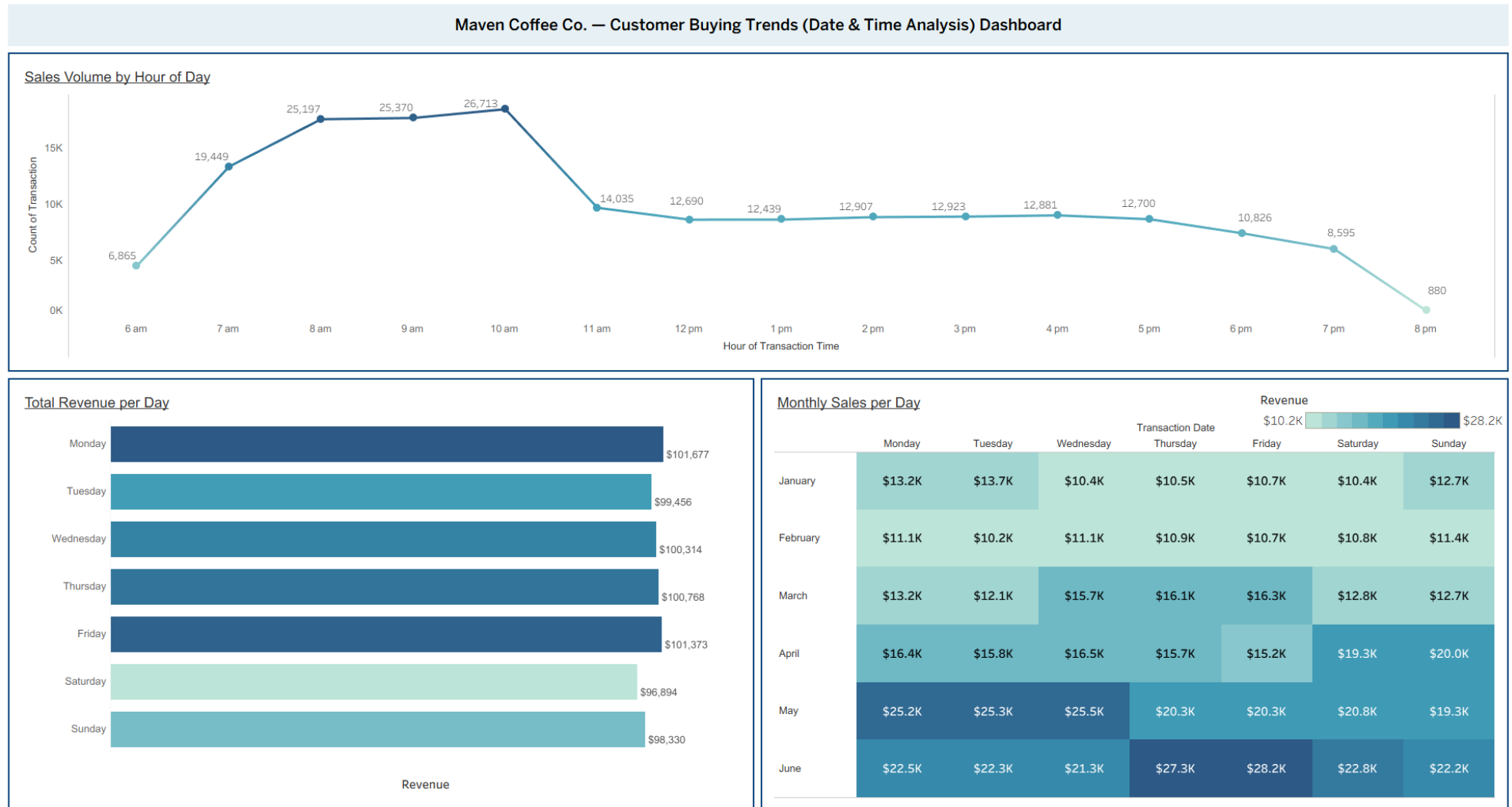


Figure 6: Dashboard 2 - Customer Trends

In Figure 6, Dashboard 2 shows customer buying pattern based on the transaction time and sales performance by day and month using a line chart, bar chart and heatmap. This dashboard helps to identify peak business hours , high-revenue days and monthly sales patterns.

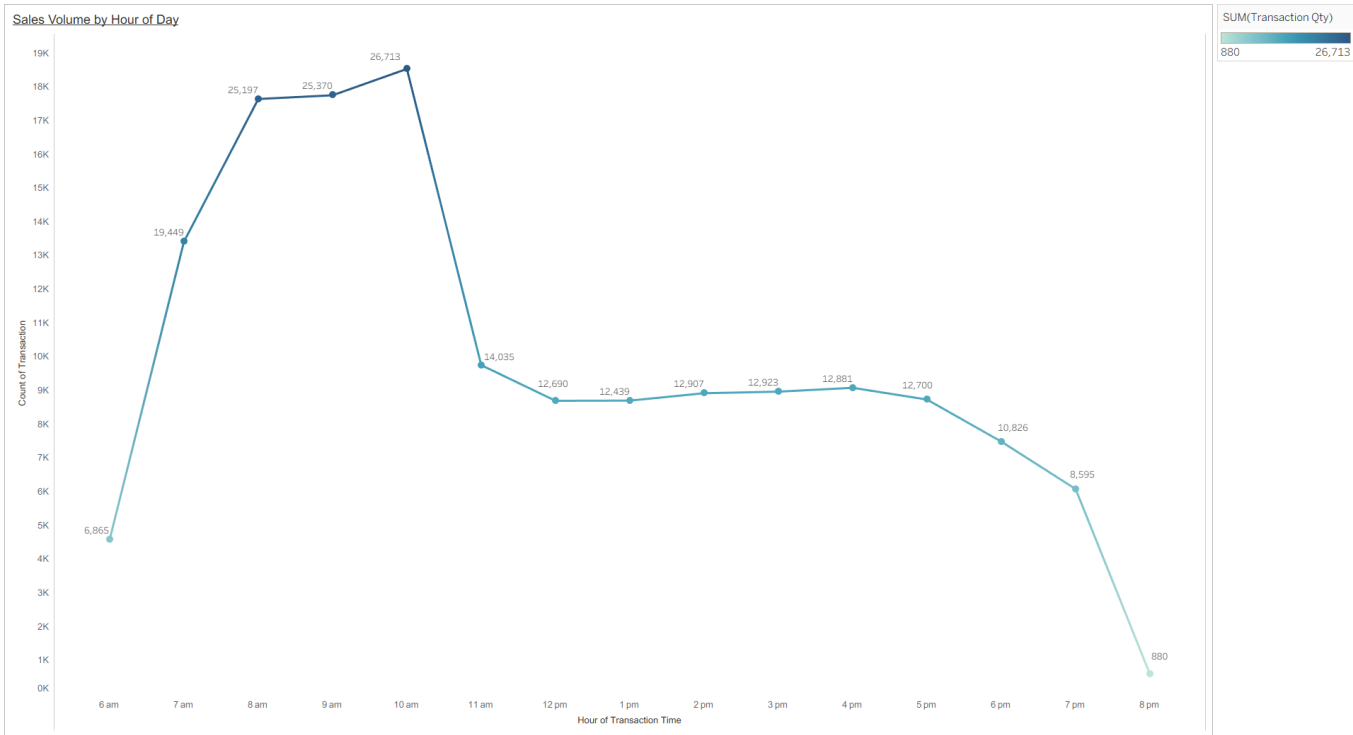


Figure 7 :Sales Volume by Hour of Day

In Figure 7, the line chart illustrates the number of transactions recorded during each hour of the day, starting from 6.00 a.m. until 8.00 p.m. The x-axis represents transaction hours while the y-axis shows transaction volume. This helps management to identify the customer purchasing activity, peak operating hours and optimise staffing schedules.

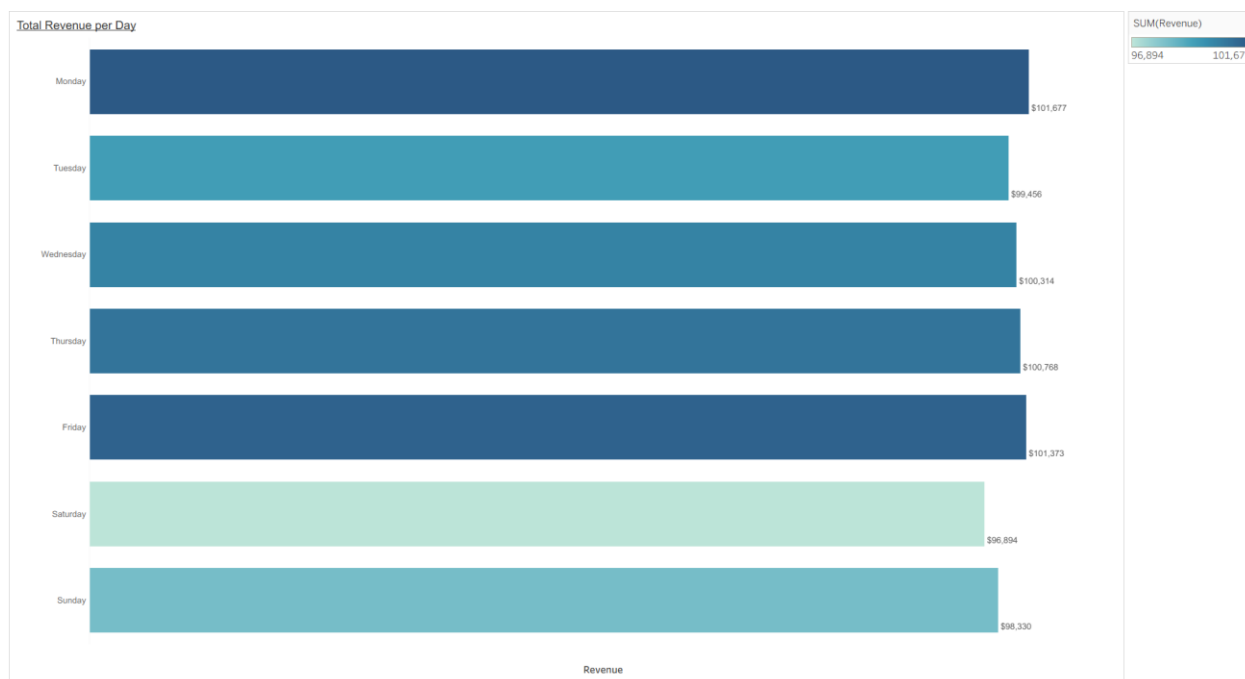


Figure 8 :Total Revenue per Day

In Figure 8, the horizontal bar chart displays the total revenue generated for each day of the week. Each bar represents the accumulated revenue for a specific day from Monday to Sunday. This helps management identify the days that contribute the most and least to overall business revenue which support better planning of promotional strategies.

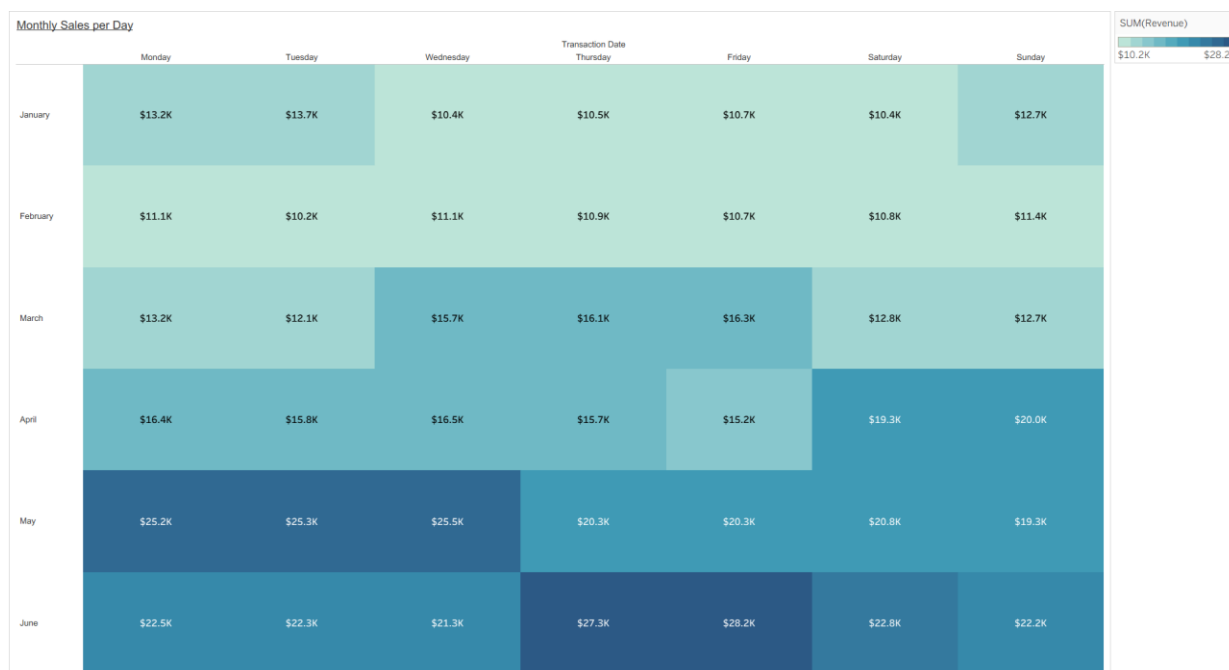


Figure 9 :Monthly Sales per Day

In Figure 9, the heatmap visualises the total revenue generated for each day of the week across different months. Darker colour intensity indicates higher revenue whereas lighter colours represent lower revenue. This helps management to identify sales patterns, high-performing periods and weaker sales periods.

### 3.3 Dashboard 3

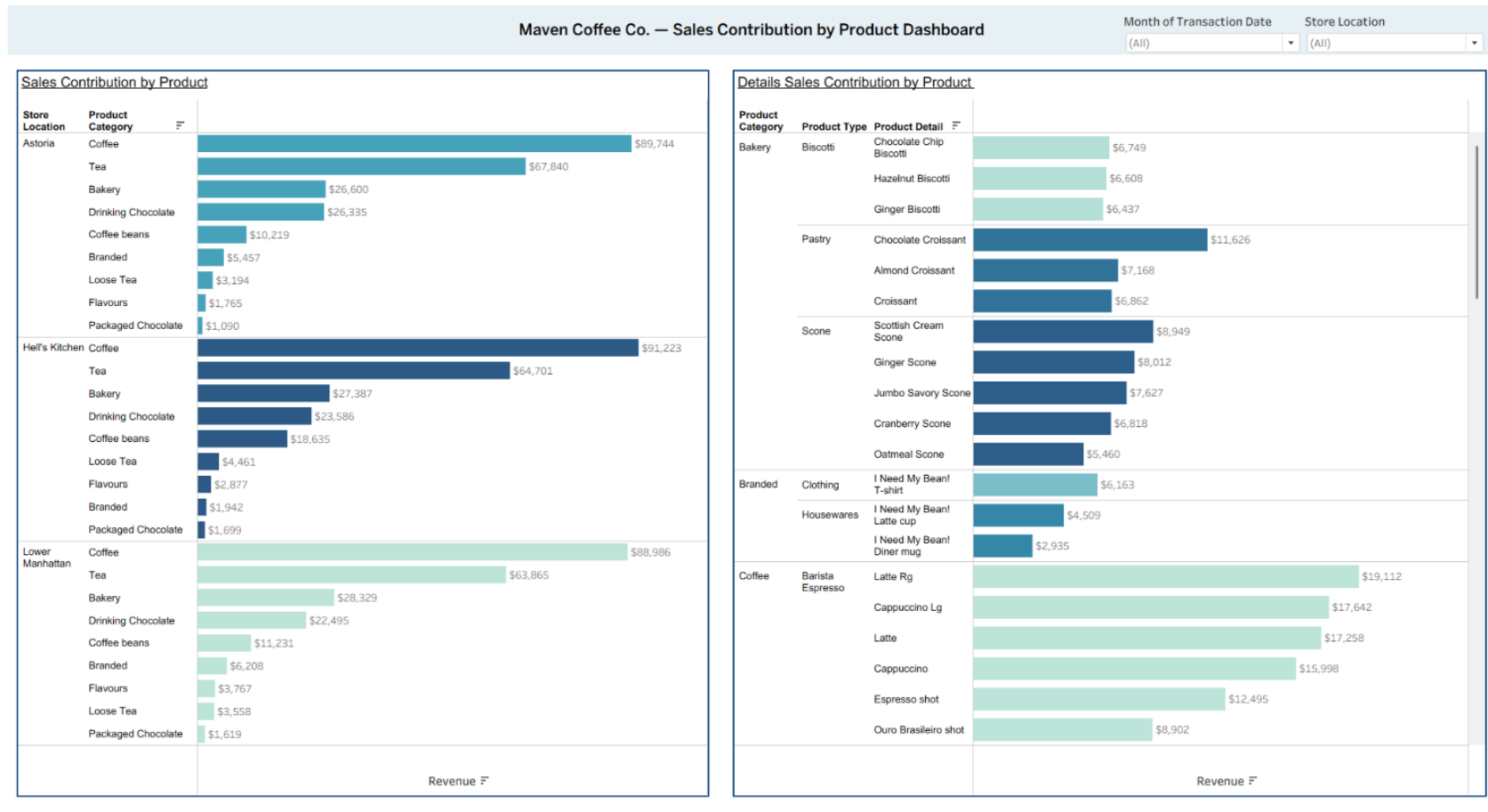


Figure 10: Dashboard 3- Interactive Sales Contribution by Product

In Figure 10, Dashboard 3 shows the sales performance by highlighting revenue across different product categories and identifying the top ten best-selling items. It also features a six-month trend line demonstrating steady revenue growth from January to June 2023, supported by interactive filters for specific dates, categories and store locations.

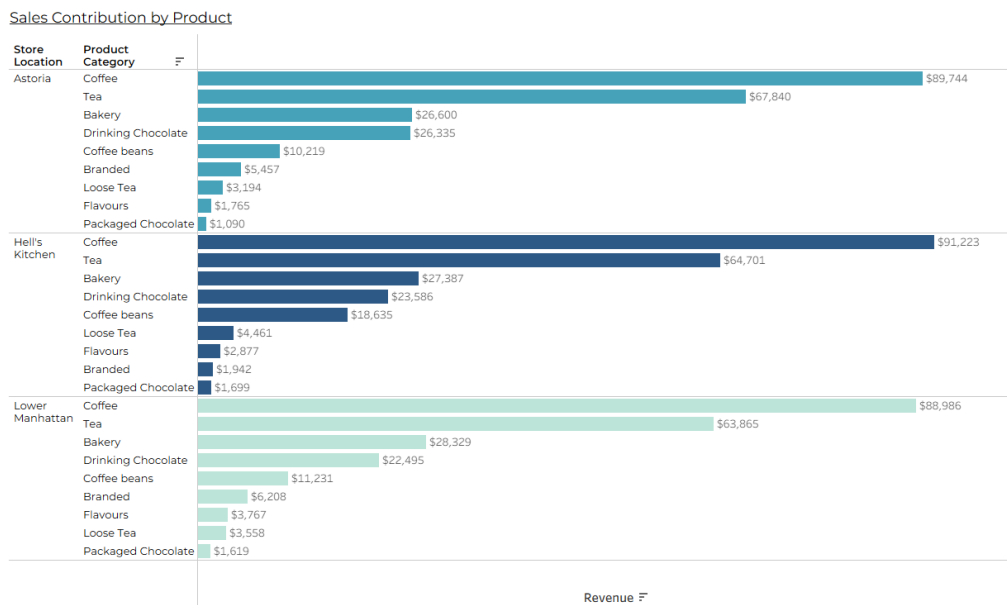


Figure 11: Sales Contribution by Product

Figure 11 shows the overall revenue earned by every main "Product Category". The data is grouped into three "Store Locations": Astoria, Hell's Kitchen, and Lower Manhattan. This graph provides a quick and high-level overview. It enables management to easily identify which general categories generate the most revenue and whether certain categories perform better in specific regions.

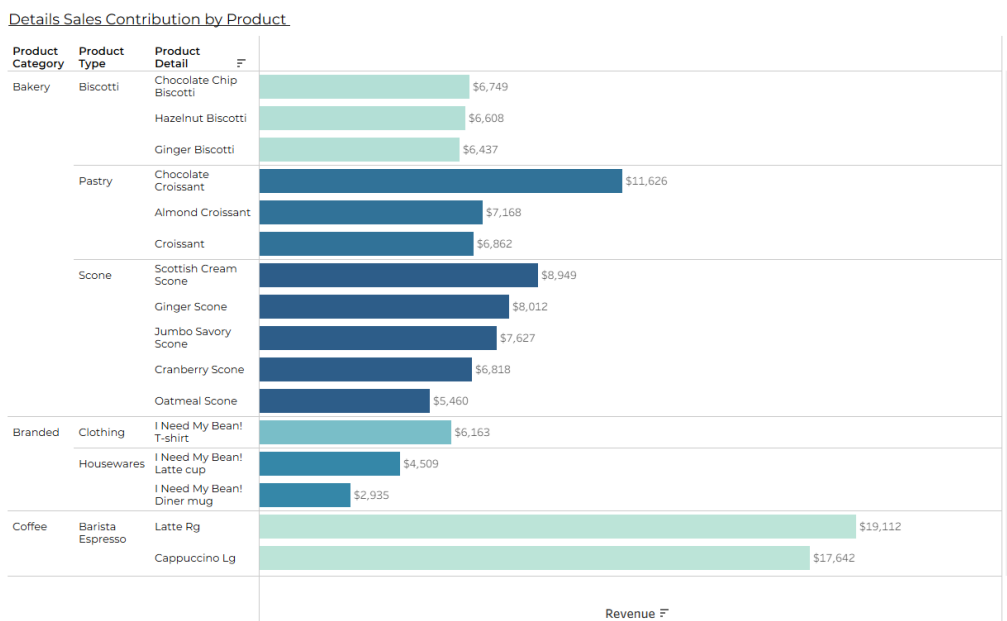


Figure 12: Details Sales Contribution by Product

Figure 12 shows graph dives further into the products. It follows a hierarchical structure of Product Category, Product Type, and Product Detail. For example, it divides "Bakery" into "Scone" and then displays the detailed sales figures for a "Scottish Cream Scone". While the first chart shows the overall picture, this chart provides the details. It is important to determine the specific best-selling goods (such as the Regular Latte) and the worst-performing items (such as Packaged Chocolate), allowing the company to make direct decisions about inventory and menu modifications.

## Advanced Tableau feature 1: Dashboard action

**Edit Filter Action**

Name: Filter1 Insert ▼

**Source Sheets**

Dashboard 3 ▼

- ☐ Details Sales Contribution by Product
- ☒ Sales Contribution by Product

**Run action on**

☐ Hover ☒ Select ☐ Menu ☐ Single-select only

**Target Sheets**

Dashboard 3 ▼

- ☒ Details Sales Contribution by Product
- ☐ Sales Contribution by Product

**Clearing the selection will**

☒ Keep filtered values ☐ Show all values ☐ Exclude all values

**Filter**

☐ All fields ☒ Selected fields

| Source Field                              | Target Data Source               | Target Field                              |
|-------------------------------------------|----------------------------------|-------------------------------------------|
| <input type="checkbox"/> Product Category | Transactions (Coffee Shop Sales) | <input type="checkbox"/> Product Category |
| <input type="checkbox"/> Click to add     |                                  |                                           |

Remove Cancel OK

Figure 13 - Dashboard Action Detail

Figure 13 shows the detail of the dashboard action feature which allows users to select a Product Category in the Sales Contribution by Product chart, and Tableau will automatically filter the Details Sales Contribution by Product chart accordingly.

## Advanced Tableau feature 2: Filters

**Month of Transaction Date** ▼

(All) ▼

Enter search text

- ☒ (All)
- ☒ January
- ☒ February
- ☒ March
- ☒ April
- ☒ May
- ☒ June

**Store Location** ▼

(All) ▼

- ☒ (All)
- ☒ Astoria
- ☒ Hell's Kitchen
- ☒ Lower Manhattan

Figure 14 - Dashboard Filters

Filters were added for Month of Transaction Date and Store Location to allow users to view sales performance based on selected months and branches as shown in Figure 14. Both filters will affect both sheets in the dashboard 3.

### Advanced Tableau feature 3: Calculated fields



*Figure 15 - Revenue Calculated Fields*

Figure 15 shows a Calculated field named Revenue which created using the formula:

$$Revenue = [Transaction Qty] * [Unit Price]$$

This calculated field is used to measure the sales value generated by each transaction and serves as a key metric in the dashboard analysis.



## 4.0 Advanced Analytics in Tableau

This section analyses data from Dashboards 1, 2, and 3 using advanced analytical techniques. The goal is to identify patterns that can guide Maven Coffee Co. in making better decisions about product focus, branch management, and daily operations.

### 1. Trend Analysis and Pattern Identification

Trend analysis was applied to examine how revenue has changed over time and to identify patterns in customer purchasing behaviour across hours and days.

- The monthly revenue shows a clear and consistent growth trend over six-month period. Total revenue increased from \$81,678 in January to \$166,486 in June which represents a growth rate of approximately 104%. A minor decline was observed in February (\$76,145) which may be caused by seasonal factors or fewer promotional activities during that period. However, sales continued to increase steadily from March to June which shows that the business is successfully expanding its customer base and increasing revenue over time.
- The Sales Volume by Hour chart reveals that sales are particularly strong in the morning. The busiest time is between 8:00 and 10:00 a.m., with a peak of 26,713 transactions at 10:00 a.m. However, after 11:00 a.m., sales drop by more than 50% and remain low throughout the afternoon. This pattern appears consistently across all six months which confirms it is a regular customer behaviour rather than a one-time event.
- The Total Revenue per Day shows that weekdays generate more revenue (averaging over \$100,000) than weekends, which fall below \$100,000. Monday has the highest daily revenue at \$101,677 while Saturday has the lowest at \$96,894. This pattern is consistent across all months which shows that weekdays reliably outperform weekends.

These patterns are important as it shows that Maven Coffee Co. follows a predictable demand cycle that directly affects how resources should be managed. The consistent morning peak and strong weekday performance suggests that most customers are working professionals who visit during their morning commute. Since the patterns are stable and predictable, it can be used to make reliable decisions about staffing and stock management. However, if this segment declines due to factors such as remote working trends or economic changes, the business has no strong afternoon or weekend revenue to fall back on. Therefore, the company should actively work to grow revenue during afternoon and weekend hours so that the business is not entirely dependent on one customer segment.

## 2. Comparative Analysis and Product Segmentation

Tableau's grouping and drill-down features were used to compare branch-level revenue performance and analyse the contribution of individual product categories, types and details to the total revenue of \$698,812.

- Interestingly, all three branches perform almost identical. Hell's Kitchen (\$236,511), Astoria (\$232,244), and Lower Manhattan (\$230,057) all have relatively similar total sales with a difference of less than 3% between the highest and lowest-performing branch. This shows that no branch is clearly performing better or worse than the others.
- Analysis of the product hierarchy from Category to Type to Detail shows few products that provide the majority of the revenue. Coffee (\$269,952) and Tea (\$196,406) are the most popular, with espresso beverages like the Regular Latte (\$19,112) and Cappuccino Large (\$17,642) being the best-sellers.
- In contrast, the absolute lowest-selling product category across all locations is Packaged Chocolate. Over the six-month period, it only earned \$1090 in Astoria, \$1699 in Hell's Kitchen, and \$1619 in Lower Manhattan. This implies that Packaged Chocolate generated less than \$5000 in total income, making it the shop's weakest product.

These patterns are important as the near-equal branch performance shows that location is not a factor in revenue differences. In addition, the large gap between Coffee and Tea products versus Packaged Chocolate shows that customer demand is strongly focused on beverages. These patterns are useful for deciding where to focus product investment and which items should be reconsidered. It also shows that Maven Coffee Co.'s best opportunity for growth is to strengthen its focus on beverages rather than expanding its current product range further.

## 5.0 Insights & Decision Support

### 5.1 Dashboard Interpretation & Discussion

The dashboard analysis provides important insights into customer behaviour, branch performance and product sales at Maven Coffee Co. The following insights directly address the core business problem of how the company should allocate its product focus, branch attention and operational strategy to improve overall sales performance.

#### **Insight 1: Peak Hour and Shift Management**

The analysis of customer purchasing patterns reveals critical timing trends that impact operational strategy is strongest during the morning period between 8.00 a.m. and 10.00 a.m. where sales volume reaches its highest point at 26,713 transactions at 10.00 a.m. However, after 11.00 a.m. transaction volume decreases significantly by more than 50% and remains low throughout the afternoon. In addition, weekday revenue consistently outperforms weekends, with average weekday revenue exceeding \$100,000 while weekends stay below this threshold. This indicates that Maven Coffee Co.'s primary customer base consists of working professionals. These findings show that operational resources such as staffing levels and stock preparation should be focused more during peak morning hours to improve service efficiency and customer satisfaction.

#### **Insight 2: Branch Attention and Marketing Strategy**

The analysis of revenue by store location shows that all three branches contribute almost equally to total revenue of \$698,812. Hell's Kitchen (\$236,511), Astoria (\$232,244), and Lower Manhattan (\$230,057) branches demonstrate balanced sales performance with less than 3% differences in revenue contribution. This indicates that no branch is significantly underperforming or dominating overall sales and that customer demand is evenly distributed geographically. Therefore, Maven Coffee Co. can apply a unified marketing strategy across all locations instead of creating separate campaigns for individual branches.

#### **Insight 3: Product Focus and Inventory Optimization**

The product analysis highlights that Coffee (\$269,952) and Tea (\$196,406) categories are the main revenue contributors collectively accounting for more than 66% of total revenue. Products such as the Regular Latte (\$19,112) and Large Cappuccino (\$17,642) are among the best-selling items which show strong and consistent customer preference for premium coffee beverages. In contrast, Packaged Chocolate records the lowest sales contribution across all branches generating \$4,408 total over the six-month period. This suggests that customer demand is concentrated on beverage products rather than packaged items. As a result, the company should focus more on expanding and promoting high performing coffee and tea products while reducing low performing inventory categories.

## 5.2 Recommendation

Based on the dashboard analysis and the key insights, the following practical business recommendations are proposed to support Maven Coffee Co. in improving its overall sales performance, operational efficiency and product strategy.

### **Recommendation 1: Optimise Staff Scheduling During Peak Hours**

Since transaction volume peaks at 26,713 transactions at 10.00 a.m., the company should assign more staff during morning shifts to ensure customers are served quickly and efficiently. The preparation tasks such as restocking supplies, brewing drinks and setting up food displays should be done before 6.00 a.m. to handle morning rush effectively. In addition, the company can reduce the number of staff on duty during low-traffic afternoon hours to cut operational costs without affecting service quality.

### **Recommendation 2: Introduce Afternoon Promotional Strategies**

The company should introduce afternoon promotions to attract customers during off-peak hours since the transaction volume declines significantly after 11.00 a.m. Strategies such as “Happy Hour” discounts between 1.00 p.m. and 4.00 p.m., bundled lunch combinations or loyalty reward multipliers after lunchtime could effectively increase customer demand and afternoon revenue contribution.

### **Recommendation 3: Implement Unified Marketing Strategy Across All Branches**

The company should implement a unified marketing strategy applicable to all branches as the revenue performance of each branch shows less than 3% difference in revenue contribution. By adopting this strategy, the company can reduce marketing costs and ensure consistent brand messaging to customers.

### **Recommendation 4: Prioritise Investment in Top Product Lines**

The company should continue investing in and promoting Coffee and Tea which contribute more than 66% of total revenue especially products such as Regular Latte (\$19,112) and Cappuccino Large (\$17,642) among the top-selling items. For example, the company can introduce seasonal variants, customised beverage options or limited-edition drinks. This will strengthen customer loyalty and help grow sales in the top-performing product categories.

### **Recommendation 5: Discontinue or Restructure Underperforming Product Categories**

The company should consider discontinuing or selling remaining stock of the Packaged Chocolate through promotional discounts as it recorded the lowest revenue contribution of \$4,408 across all three branches. The freed shelf space and inventory budget can then be redirected towards high-demand products like Coffee and Tea which can improve both operational efficiency and overall profitability.

In conclusion, Maven Coffee Co. should prioritise a beverage-focused growth strategy supported by operational optimisation and demand-based resource planning. These five strategies can help improve service, reduce unnecessary operational and marketing costs and increase overall sales performance across all branches.